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Data Mining

How to get useful information from data? 





Hi!

My name is Antonina Bondarchuk, I hold a Technical Trainer position at Grid Dynamics.

My specializations are:
Python, Data Science and
Big Data Engineering.

Contact me:
antonina.bondarchuk1@gmail.com



Agenda



01

Definition

Goal of
Data Mining

02

CRISP-DM

Steps discussion through
the example

03

Application

Main direction where ML
algorithms are applicable

04

Spheres

Real-life spheres to apply
ML algorithms

05

Example

Description of Medical
System



01



Definition

What is Data Mining?
What is it dedicated to?
Why is it important to
solve business problems?

Data Mining overview

Definition

Data mining, also known as knowledge discovery in data, is the process of uncovering patterns and other valuable information from large data sets. [1]

Problems

- There is data, but no outcomes.
- How to use this data?
- Is there something interesting?
- Can we get business value?

Solutions

- Techniques to find insights.
- Works even with large datasets.
- Answers business questions.
- Standardized process.



02



CRISP-DM

Cross-industry standard
process for data mining



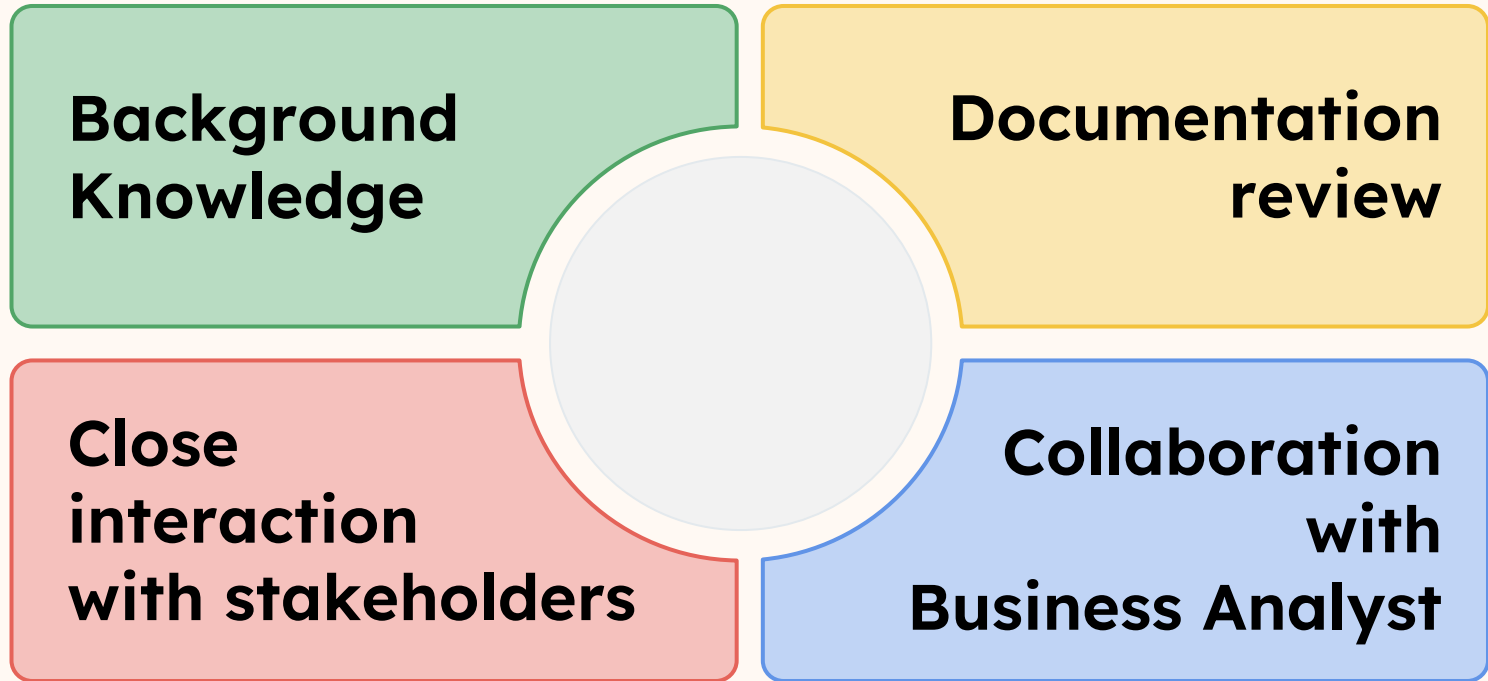
CRISP-DM diagram



I. Business Understanding

Before even take look at data we have to fully realize how processes work.

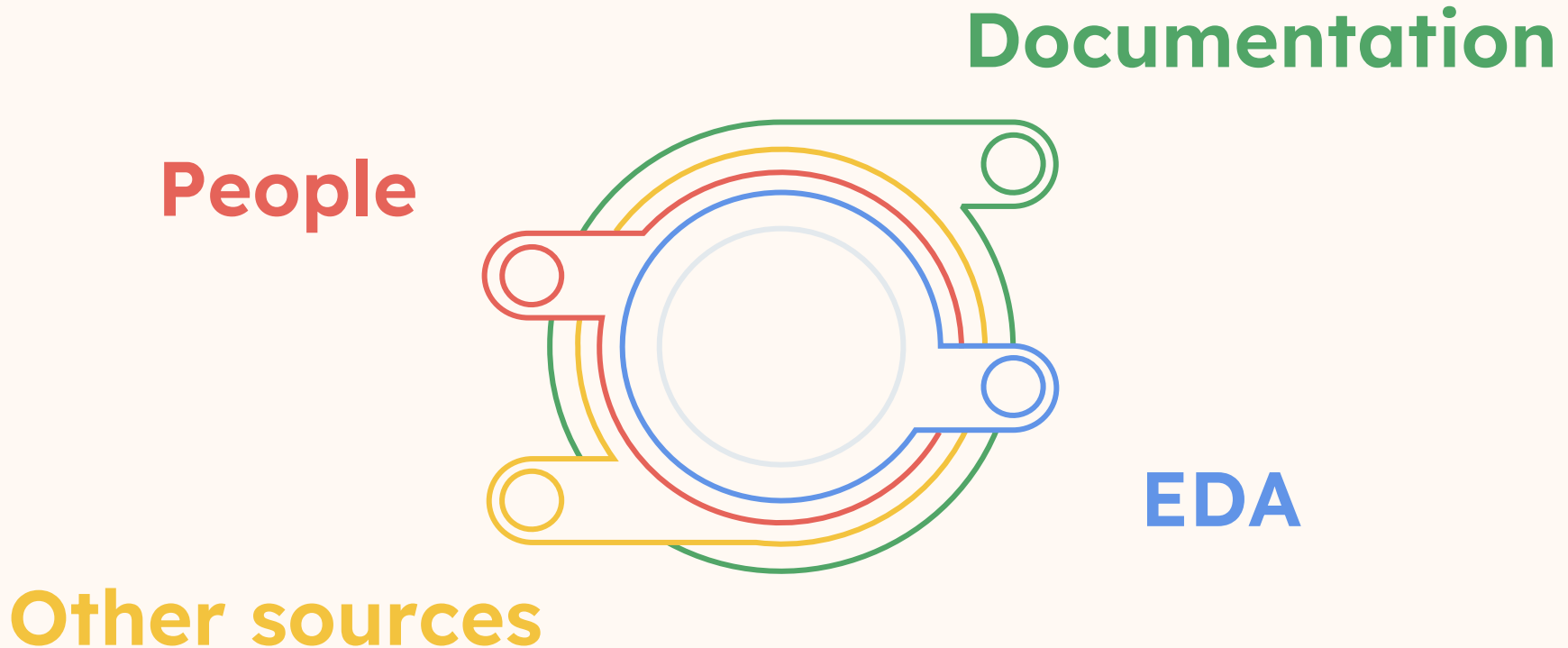
Main components of business understanding



II. Data Understanding

What do we have, how is it stored and what is inside?

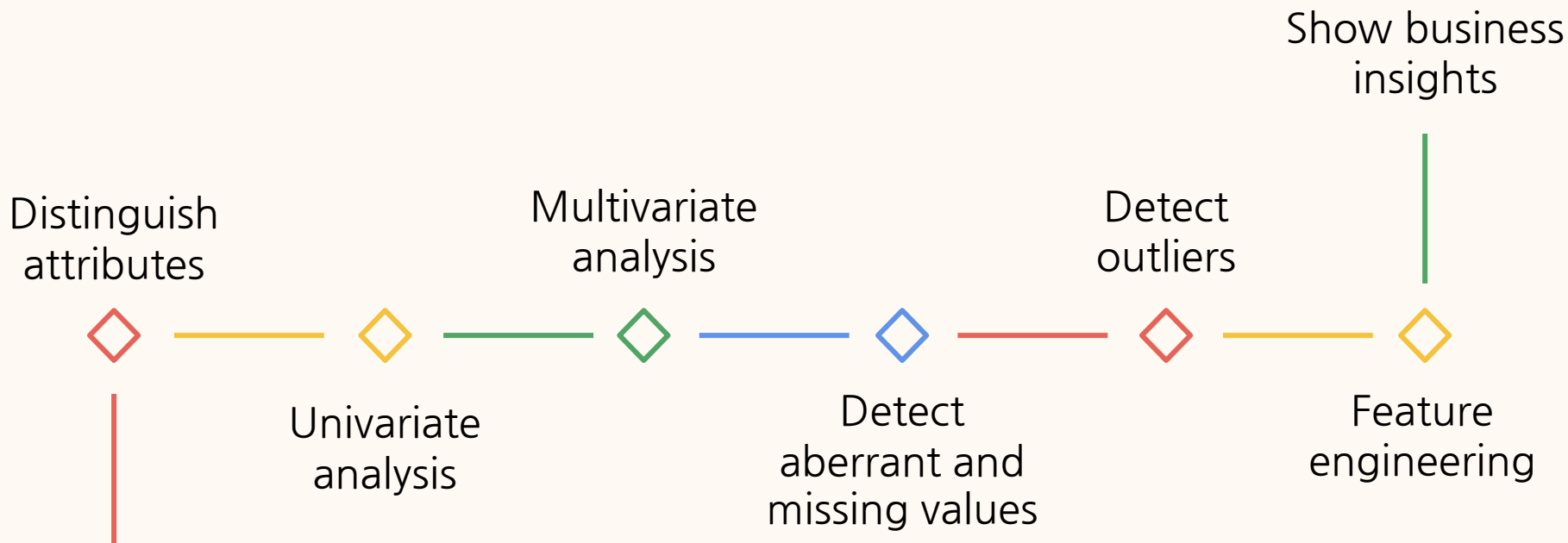
Sources to collect information about data



Exploratory Data Analysis (EDA)

“garbage in, garbage out”

[5]



Load data

[Bakery Data Analysis Example](#)

III. Data Processing

Data is already explored, now it is time to act. Let's prepare it for modeling.

Common data preparation stages



Missing Data

Filling with statistics or removing a column



Outliers

Removing them or transforming a feature



Misspelling

Using correcting algorithms



Disbalanced data

Choosing a sample and/or a proper algorithm



Duplicates

Cleaning data from duplicates



Personal data

Validating data ethically and legally

IV. Modeling

Split the data. Train the model.

Programming algorithms' template

INPUT

$X_1 = 16$
 $X_2 = 144$
 $X_3 = 49$

WHITE BOX

$Y = \text{sqrt}(X)$

OUTPUT

$Y - ?$

ML algorithms' template

INPUT

$X_1 = 16$
 $X_2 = 144$
 $X_3 = 49$

BLACK BOX

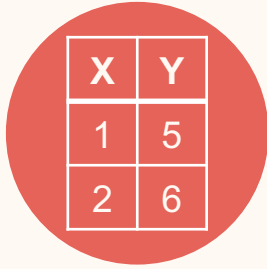
OUTPUT

$Y_1 = 4$
 $Y_2 = 12$
 $Y_3 = 7$

$y^*: X \rightarrow Y$?

Types of learning ML algorithms

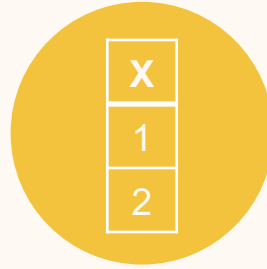
by feedback types



X	Y
1	5
2	6

Learns from labelled examples, i.e. desired output

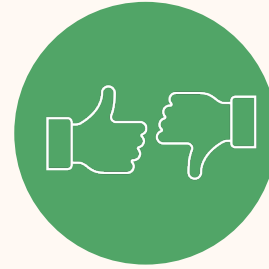
Supervised



X
1
2

Learns from unlabelled examples, desired outcomes are unknown

Unsupervised



Learns by taking actions. Requires the rewards or punishment

Reinforcement

Supervised learning. Example.

X	Y
1	5
2	6

The next faced picture will contain hot-dog or not?

X



Y

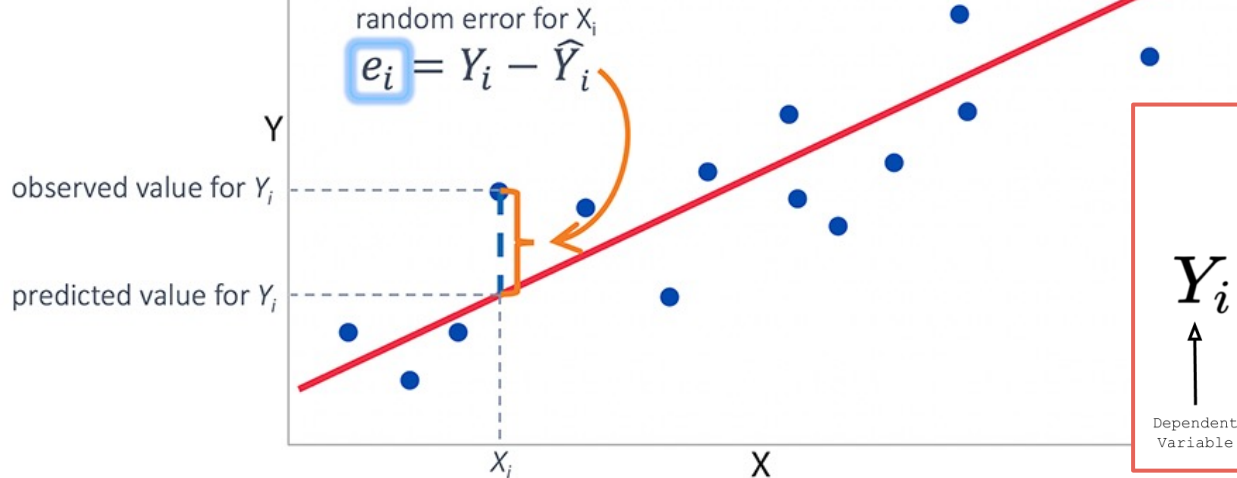
Hot-dog

NOT Hot-dog

Linear Regression

Method of Least Squares

$$\sum e_i^2 = \sum (Y_i - \hat{Y}_i)^2$$



$$Y_i = \beta_0 + \beta_1 X_i$$

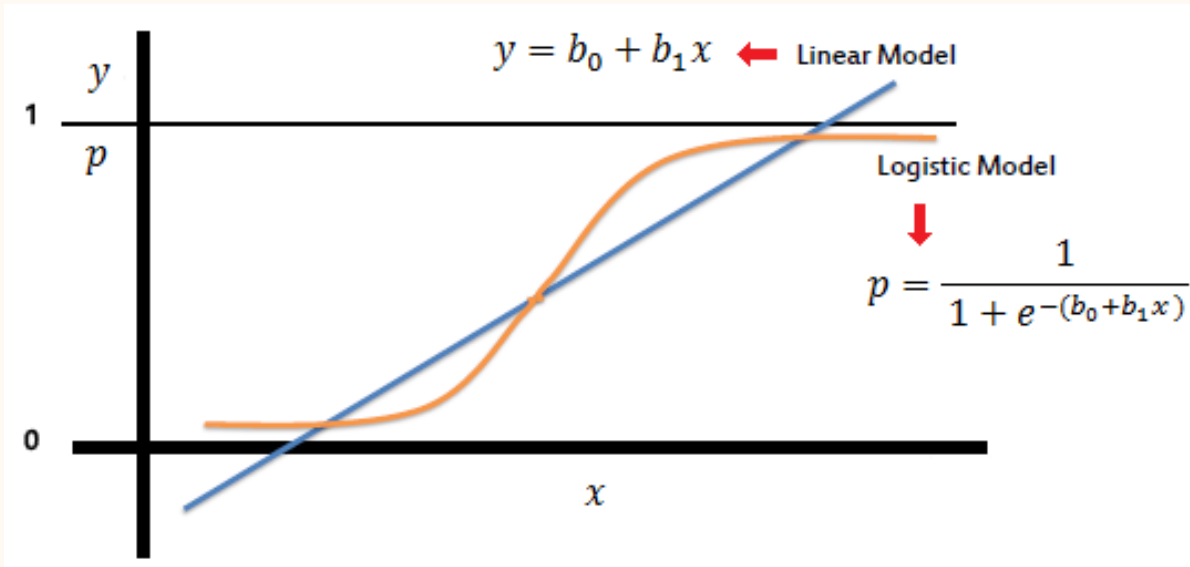
Constant/Intercept

Independent Variable

Dependent Variable

Slope/Coefficient

Logistic Regression. Classification I



The model works by predicting the probability that Y belongs to a particular category **by first fitting the data to a linear regression model**, which is then passed to the logistic function. The logistic function will always produce a S-shaped (**Sigmoid**) curve, so regardless of X , we can always obtain a sensible answer (between 0 and 1). If the **probability** is above a certain predetermined threshold (e.g. $P(\text{Yes}) > 0.5$), then the model will predict Yes.

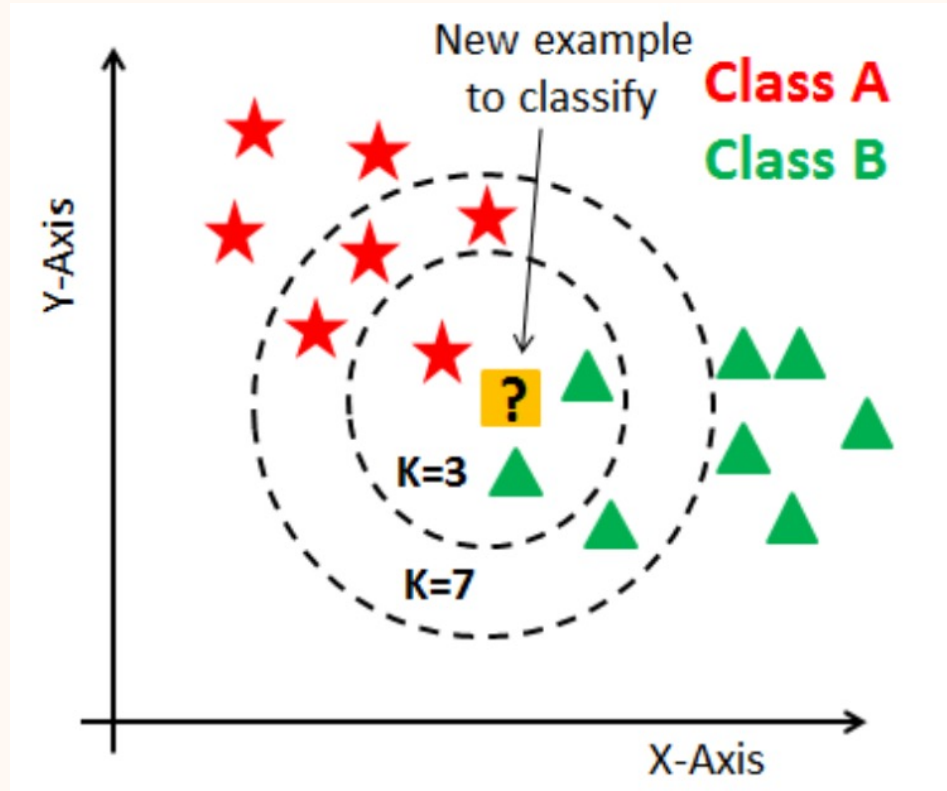
k-NN. Classification II

K-Nearest Neighbors Algorithm:

» Compute distance $D(x, p_i)$ from point x to all points.

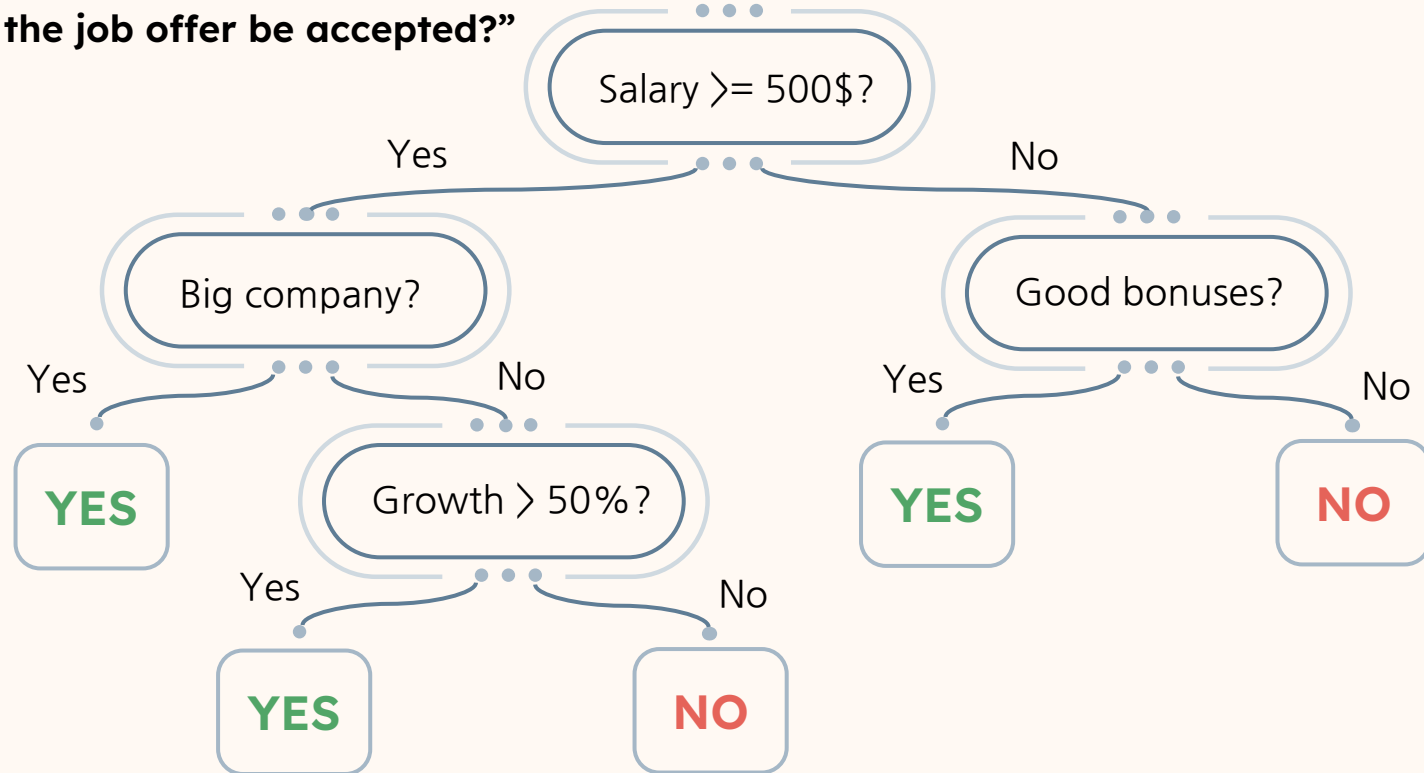
» Select k closest points and their labels.

» Output class with most frequent labels in k points.



Decision Tree. Classification III

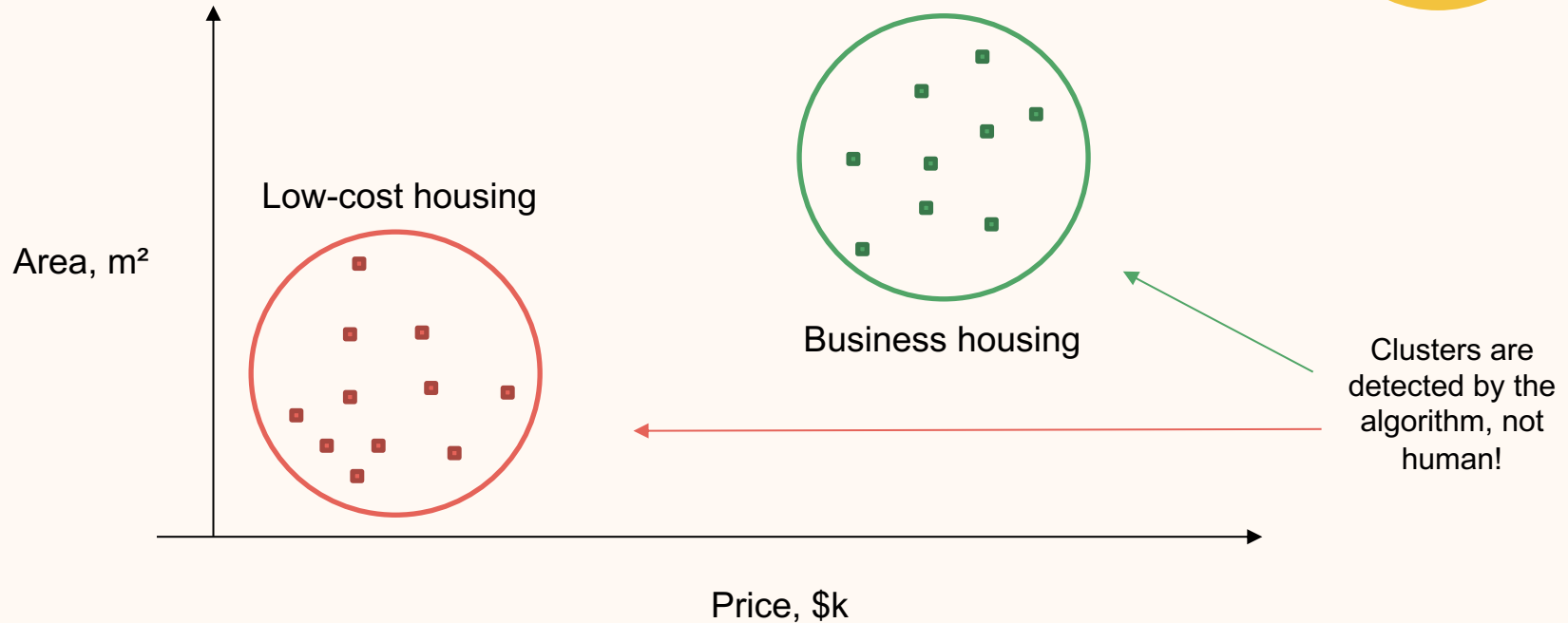
“Should the job offer be accepted?”



Unsupervised learning. Example.

x
1
2

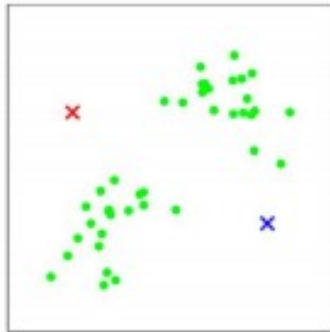
The next faced flat will be closer to business or low-cost segment?



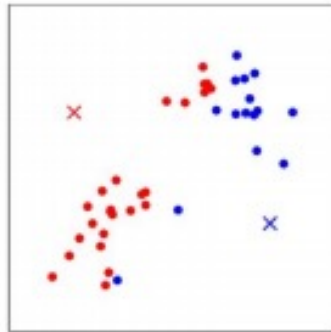
K-means. Clustering II



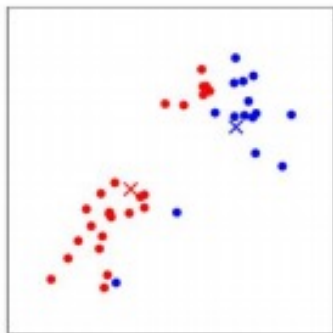
(a)



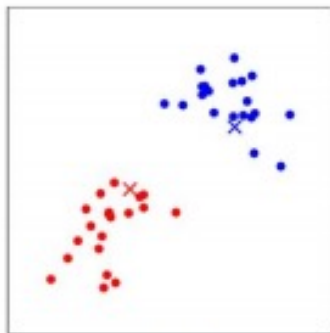
(b)



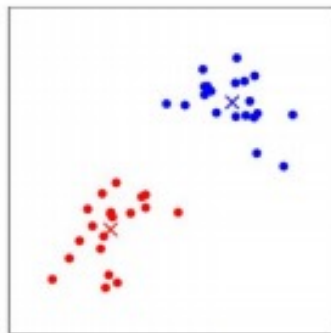
(c)



(d)



(e)



(f)

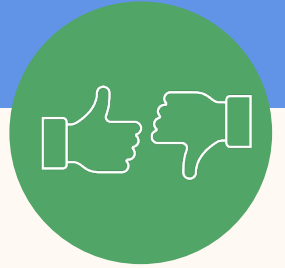
(a) Original dataset.
(b) **Random initial** cluster centroids.
(c-f) Illustration of running two iterations of k-means.

In each iteration, we assign each training example to **the closest cluster centroid** (shown by "painting" the training examples the same color as the cluster centroid to which is assigned); then we **move each cluster centroid to the mean** of the points assigned to it.

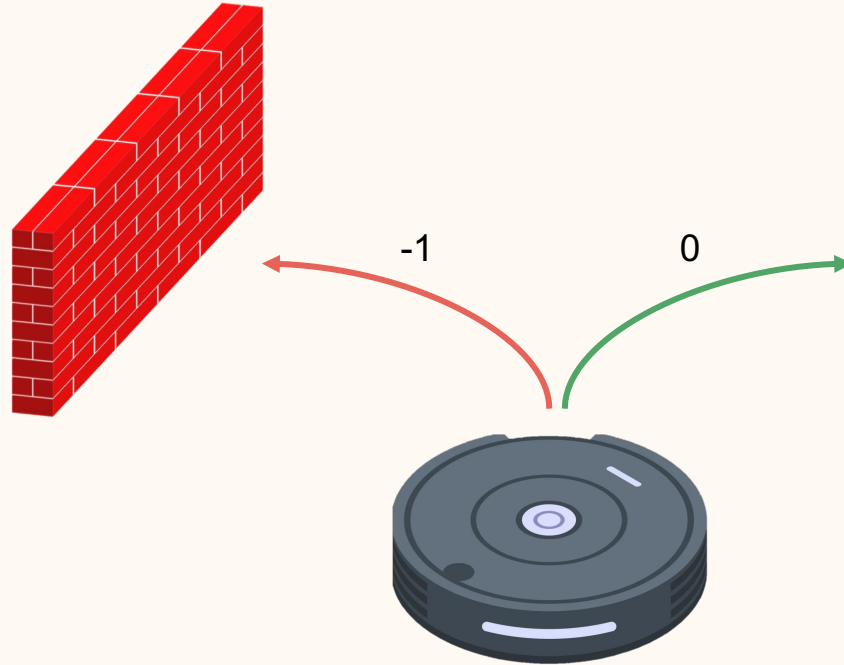
Classification VS Clustering

	Classification	Clustering
Classes	Predefined, known in advance	Undefined, unknown in advance
Examples of algorithms	Decision Tree, Logistic Regression, k-NN	k-means and its updated versions, DBSCAN

Reinforcement learning. Example.



Where must
robot vacuum
cleaner wash
and where
must not?



V. Evaluation

Do we have satisfying result? How to calculate it?

Common evaluation metrics (Supervised Learning)

Confusion matrix

		<u>Expectation</u>	
<u>Reality</u>		TRUE	FALSE
	TRUE	TP	FP
	FALSE	FN	TN

 **Accuracy**

 **F1-Score**

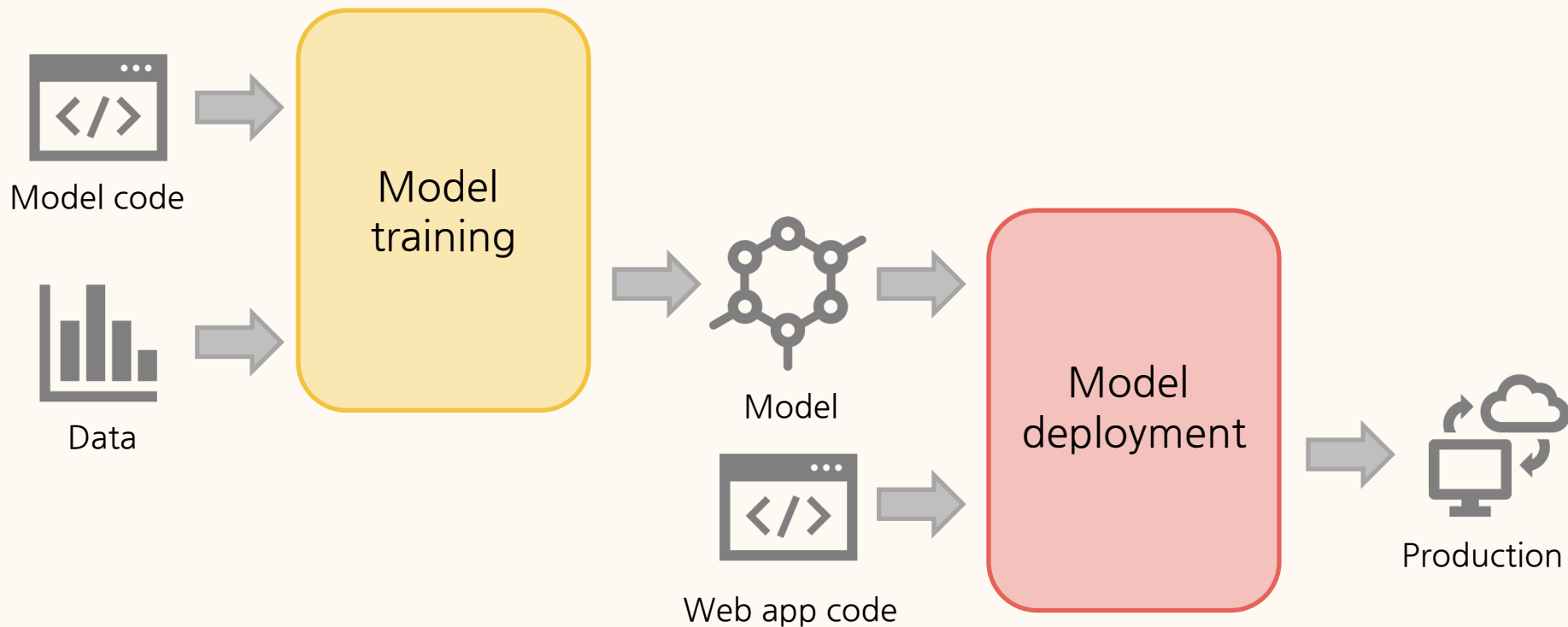
 **Sensitivity, specificity**

 **Area Under ROC curve**

VI. Deployment

Connect with other systems so that absolute beginners will be able to make value out of data.

General deployment schema



CRISP-DM example. UkrPoshta

1

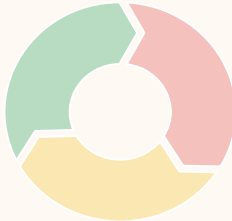
Loss of
standing clients

2

Data: Where?
How? What?

6

API
Integration



3

Fix data and
prepare

5

Score and
confusion matrix

4

Clustering and
regression



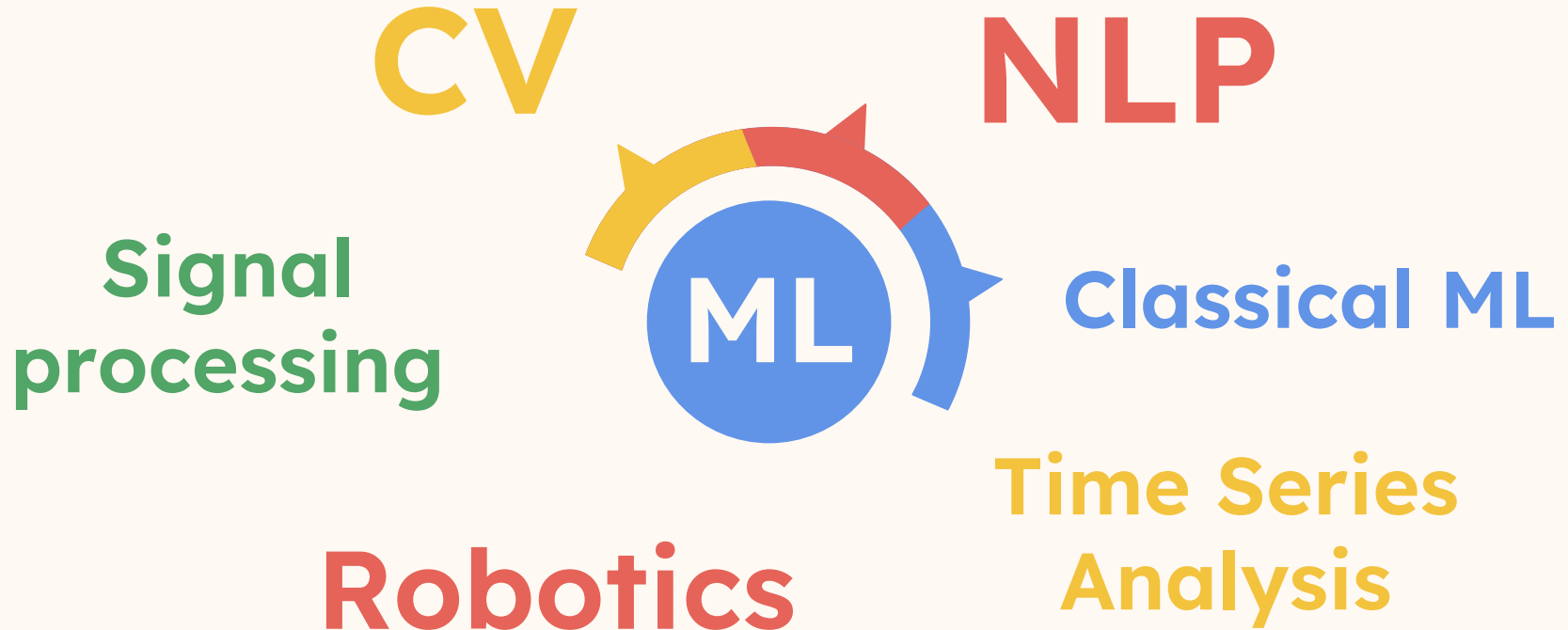
Application

Main directions of
Machine Learning work

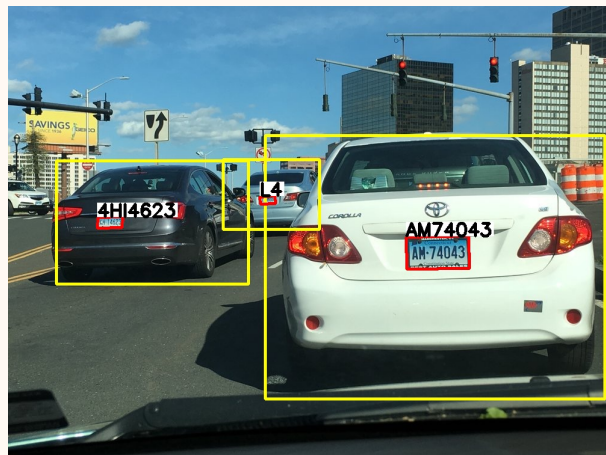
03



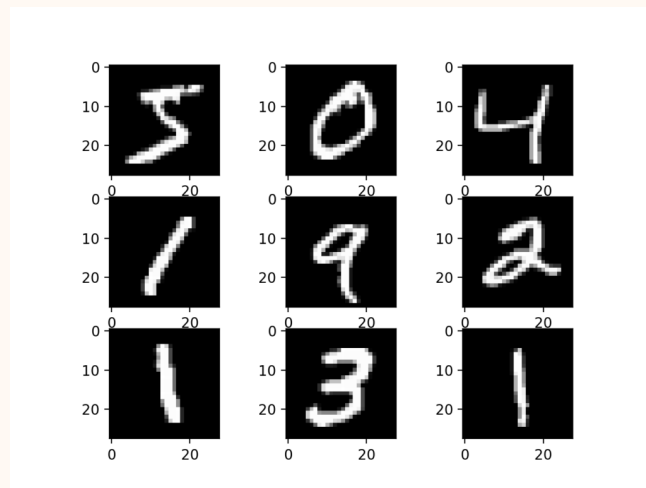
Main directions of ML/DS work



Computer Vision (CV). Examples.



Auto plates detection and recognition

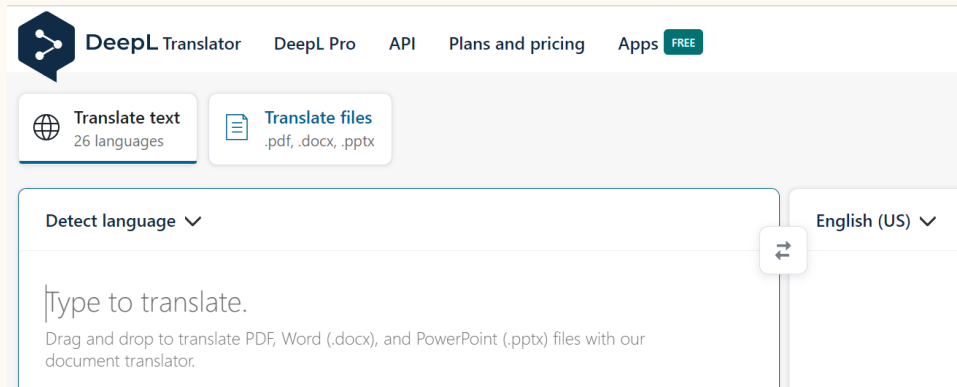


Hand-written numbers classification

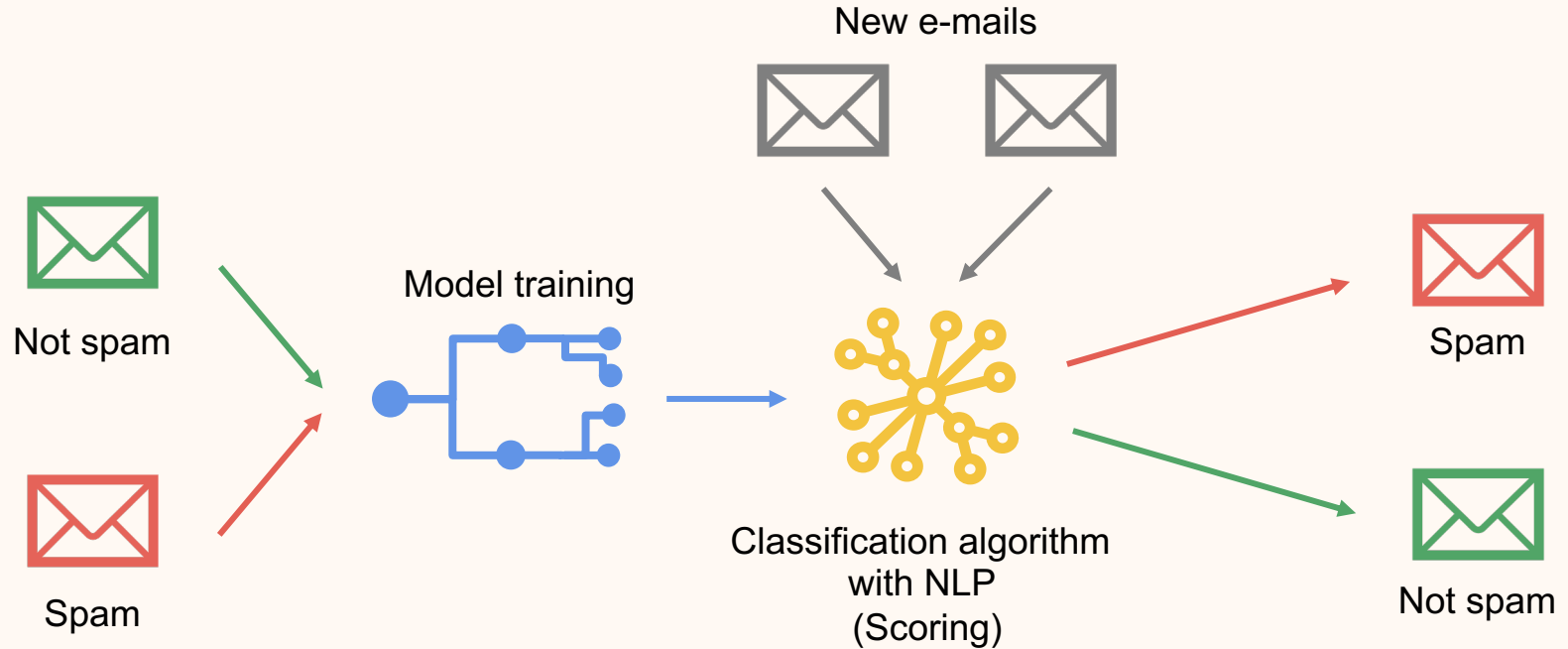
Natural Language Processing (NLP)

Me: Siri remind me to dye my hair tomorrow please.

Siri next morning:



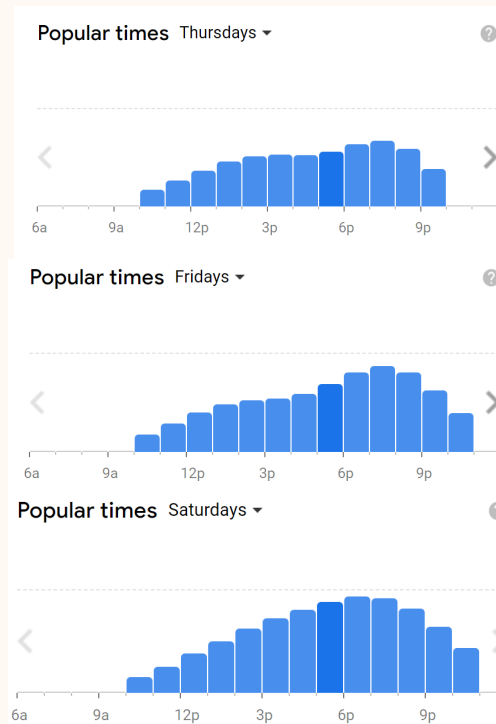
Machine Learning in Spam Filtering



Time Series Analysis. Examples.

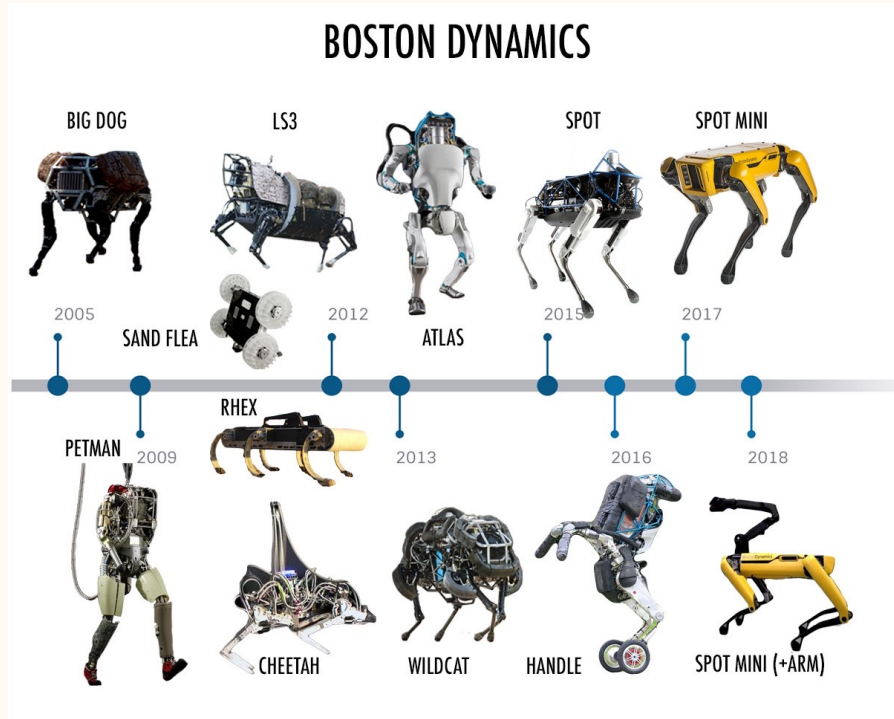


USD to UAH exchange rate
during the last 5 years



Popular times (Thu-Sat) at Yaposhka

ML in Robotics. Examples.



Signal processing. Examples.



Noisy audio

Noise reduction
algorithm



Clear audio

04



Spheres

Medicine, Finances,
Governmental sector.
Where else ML can be used?





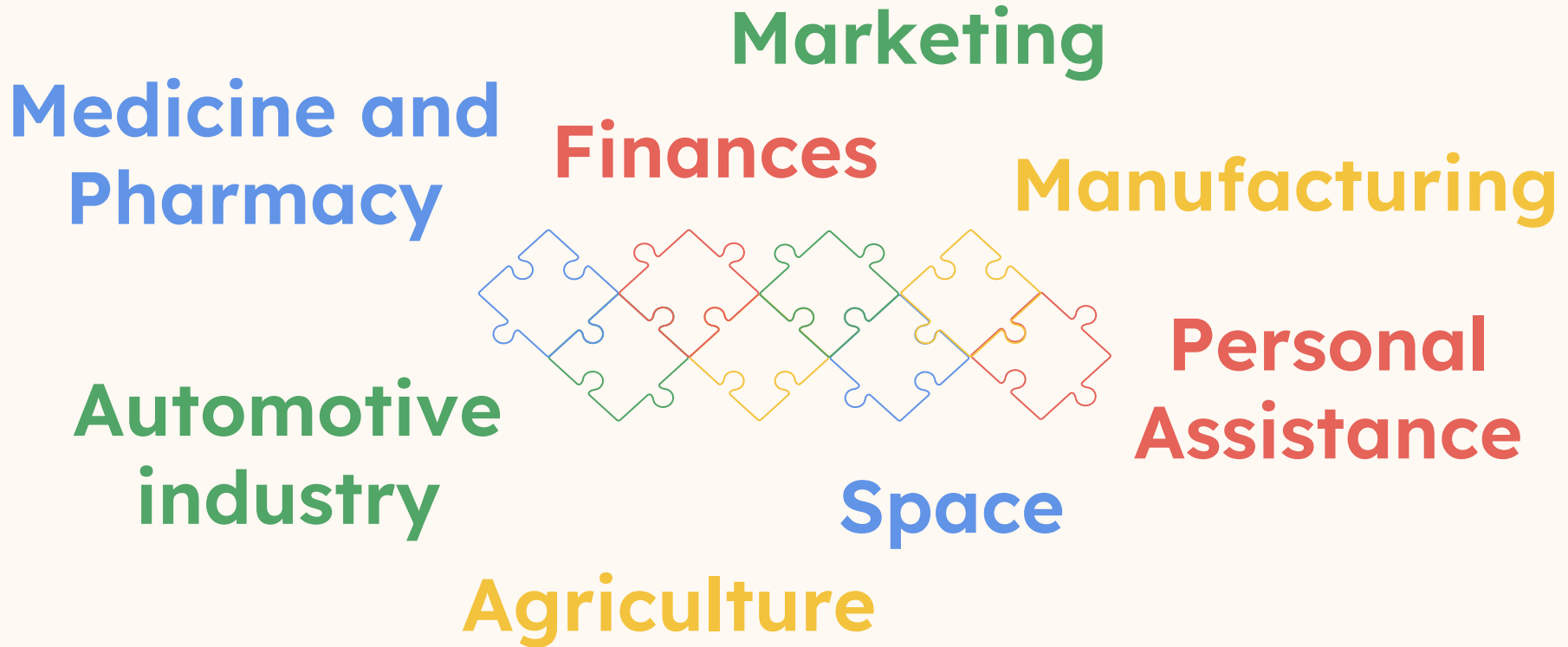
12,000

Careers are now in the world



<https://www.careerplanner.com/ListOfCareers.cfm#:~:text=List%20of%20over%2012%2C000%20Careers>

Main spheres of ML application



ML in Finances

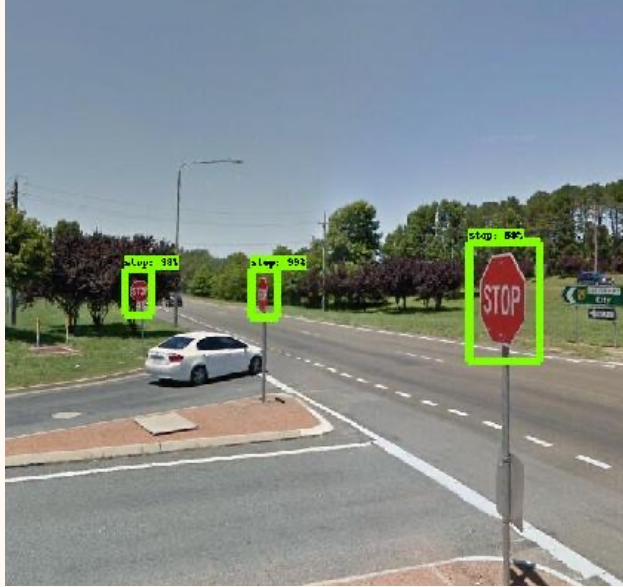


Credit card fraud detection



Crypto currency prediction

ML in Automotive Industry

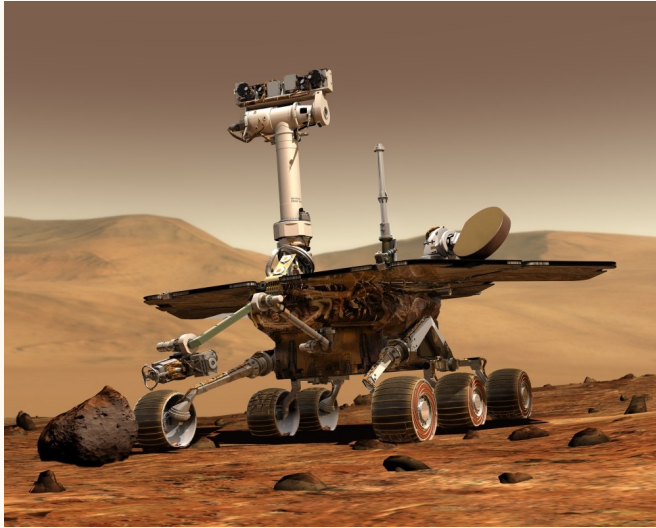


Road signs detection and recognition

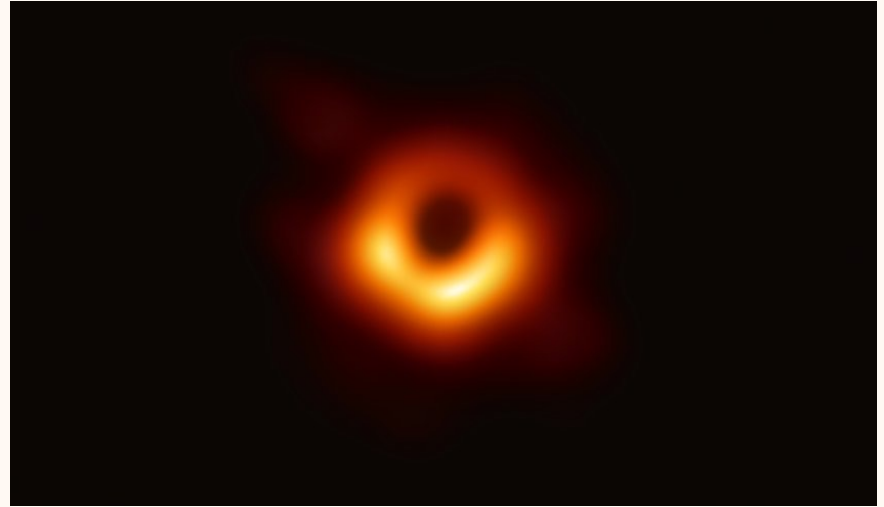


Tesla's autopilot interface

ML in Space

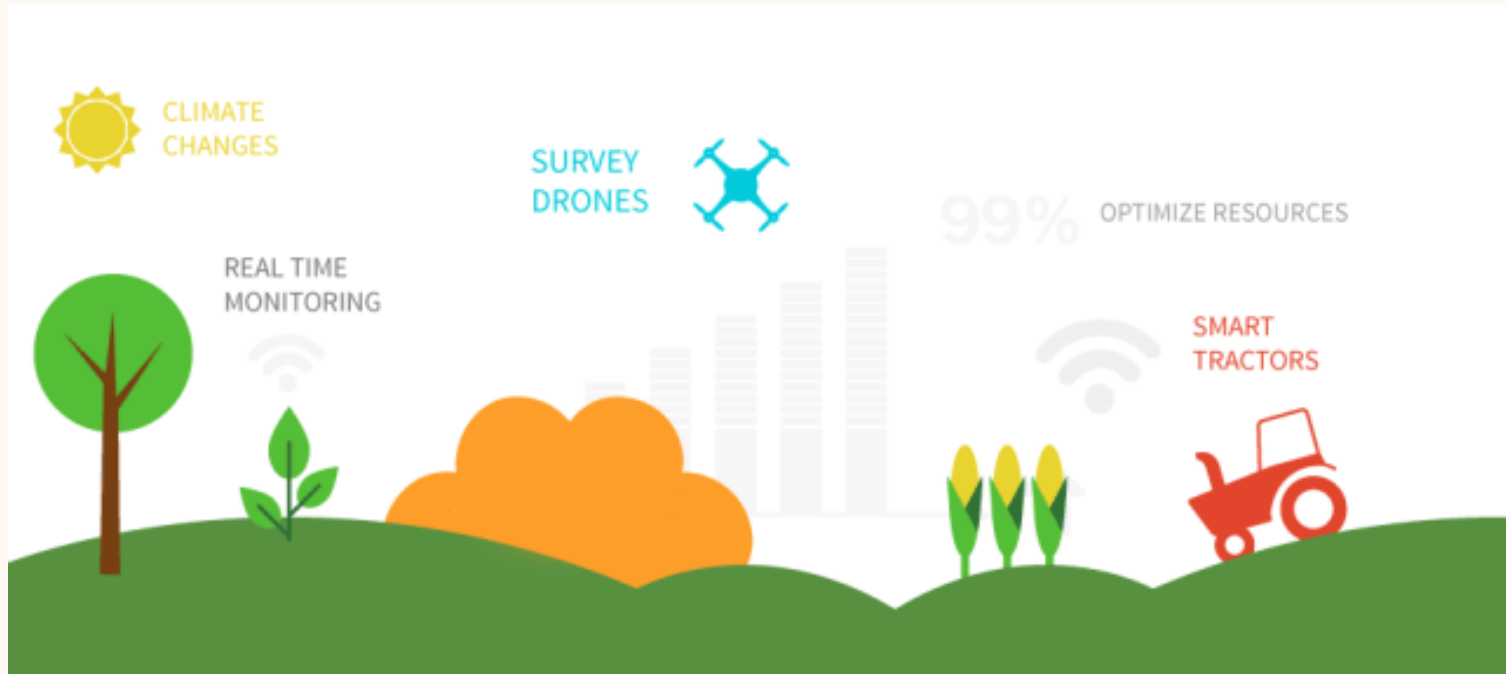


NASA's autonomous rovers (robots) that avoid obstacles on the rough surface (currently the surface of Mars)

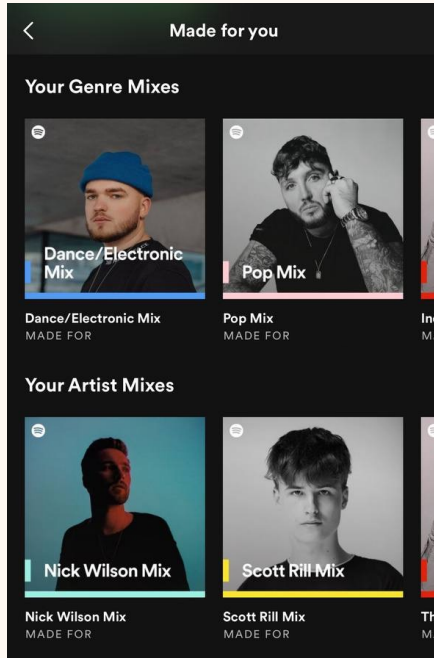


First Image of Black Hole from the Event Horizon Telescopes which was too large but became possible to process with ML algorithm

ML in Agriculture



ML in Personal Assistance



Spotify recommendations



Apple Watch Fall detection

ML in Manufacturing and Sales



Quality assurance with video analytics



System of temperature control

ML in Marketing



Personalized ads (e.g. Google Ads)



Analysis of customers' journeys
in the supermarkets



Example

Description of Risk of
Hospitalization Service



05

The Watson Health ROH Models

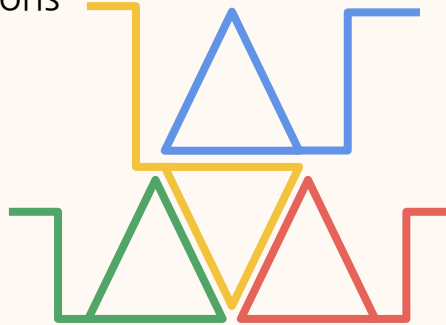
... are specifically tailored to patients with four common chronic conditions: asthma, diabetes, congestive heart failure (CHF) and chronic obstructive pulmonary disease (COPD).

Predicts an estimate of the **likelihood of hospitalization** for these conditions

Calculates risk of all-cause, related and potentially avoidable **admissions**

Estimates of the patient's risk of hospitalization within **one, three and six months** in the future

Contains model **variables** which have the greatest impact on the patient's risk score



ROH Variables Characteristics



Demographic

age, gender, number of months enrolled, mental health coverage



Inpatient stay

counts, types, recency, duration and payment amounts



Outpatient visit

counts, recency and payment amounts, counts of procedure groups



Prescription drug

Medication possession ratios, recency, with breakouts for chronic and acute drugs



Disease history

Counts of distinct Watson Health Disease Staging categories, with breakouts by disease stage

Sample output of the Watson Health ROH for a diabetes patient



**Patient A – diabetes admission
type 1 – potentially avoidable**

Month 1 score: 0.486

Month 3 score: 0.648

Month 6 score: 0.776

Most significant risk drivers	Value
Average count of ER visits over enrolled months	1.18
Maximum disease stage for target disease	3.06
Average count of inpatient medical stays over enrolled months	0.82
Average inpatient days over enrolled months	3.18

Thanks!

Do you have any questions?

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References

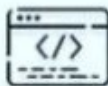
1. <https://www.ibm.com/cloud/learn/data-mining>
2. <https://www.tableau.com/learn/articles/what-is-data-mining>
3. https://drive.google.com/file/d/1se-WUIJ_kDYk0AnqBxm434c49vfbEE-1/view?usp=sharing
4. <https://towardsdatascience.com/an-extensive-guide-to-exploratory-data-analysis-ddd99a03199e>
5. https://www.researchgate.net/figure/Exploratory-Data-Analysis-EDA-steps_fig2_342282008
6. <https://stanford.edu/~cpiech/cs221/handouts/kmeans.html>
7. https://www.jmp.com/en_hk/statistics-knowledge-portal/what-is-multiple-regression/fitting-multiple-regression-model.html

Data Scientist



uses statistics and machine learning to make predictions and answer key business questions

Skills - Math, Programming, Statistics

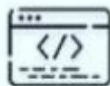


Data Engineer



build and optimize the systems that allow data scientists and analysts to perform their work

Skills - Programming, BigData & Cloud



Data Analyst



deliver value by taking data, communicating the results to help make business decisions

Skills - Communication, Business Knowledge

